AMAZON Delivery *Package Pick time Efficiency*

Package Types

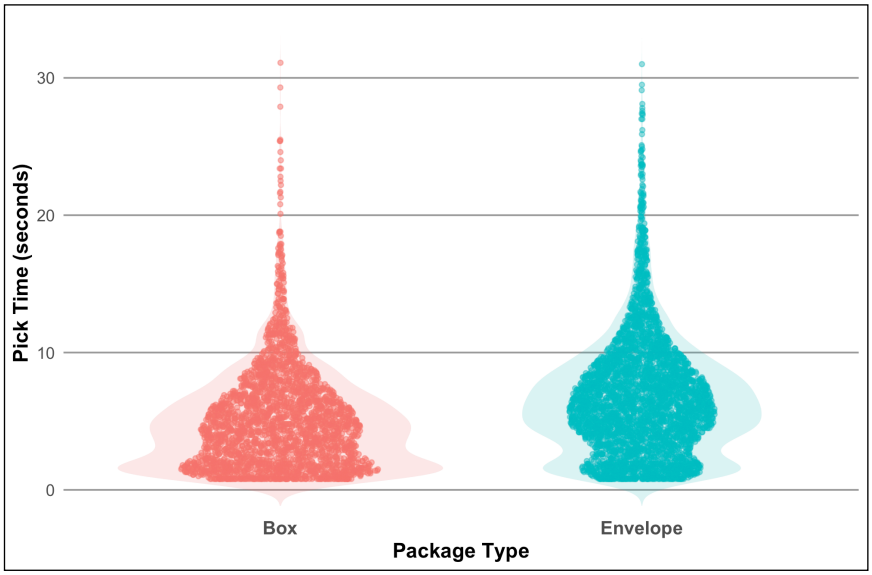
Amazon delivery routes contain two primary package types: **boxes** and **envelopes**. // Boxes are placed individually in delivery bags, making them easier to see and grab. Envelopes are stacked in groups, often requiring flipping through multiple packages to locate the correct one. // Because of this difference in organization, envelopes typically take longer to find than boxes. Understanding this sorting system is essential for interpreting the pick-time data shown throughout this report.

Overall Month Stats

Metric	Box	Envelope
Mean Pick Time (s)	4.2	5.6
Median Pick Time (s)	3.7	5.0
Standard Deviation	2.9	3.8
90th Percentile	8.7	11.4

Figure 1. Distribution of Pick Times by Package Type

This violin plot shows the distribution of search times for boxes and envelopes across the month. Individual points represent each package search. Boxes tend to have shorter pick times because they are sorted individually, while envelopes are stacked together, making them harder to locate quickly. This package-type contrast carries forward into **Figure 2**, **Figure 4**, and **Figure 7**.



Comparing Weekly Efficiency

As drivers gain familiarity with their organization system, search times can improve. This section compares the distribution of package pick times across four weeks to observe whether efficiency improves over time.



Figure 2. Weekly Distribution of Pick Times by Package Type

This figure shows mirrored histograms comparing pick-time distributions for boxes and envelopes across four weeks. Each panel represents one week. Over time, the distributions shift toward lower pick times, suggesting improved efficiency as the organization system becomes more familiar. This weekly improvement provides the broader time-based context for the daily performance shown later in **Figure 3**.

Building Trends, Day by Day

The following charts examine how package search patterns develop throughout a route. By looking at the data day by day as new bags are opened, broader trends in organization, package type, and search time begin to emerge.

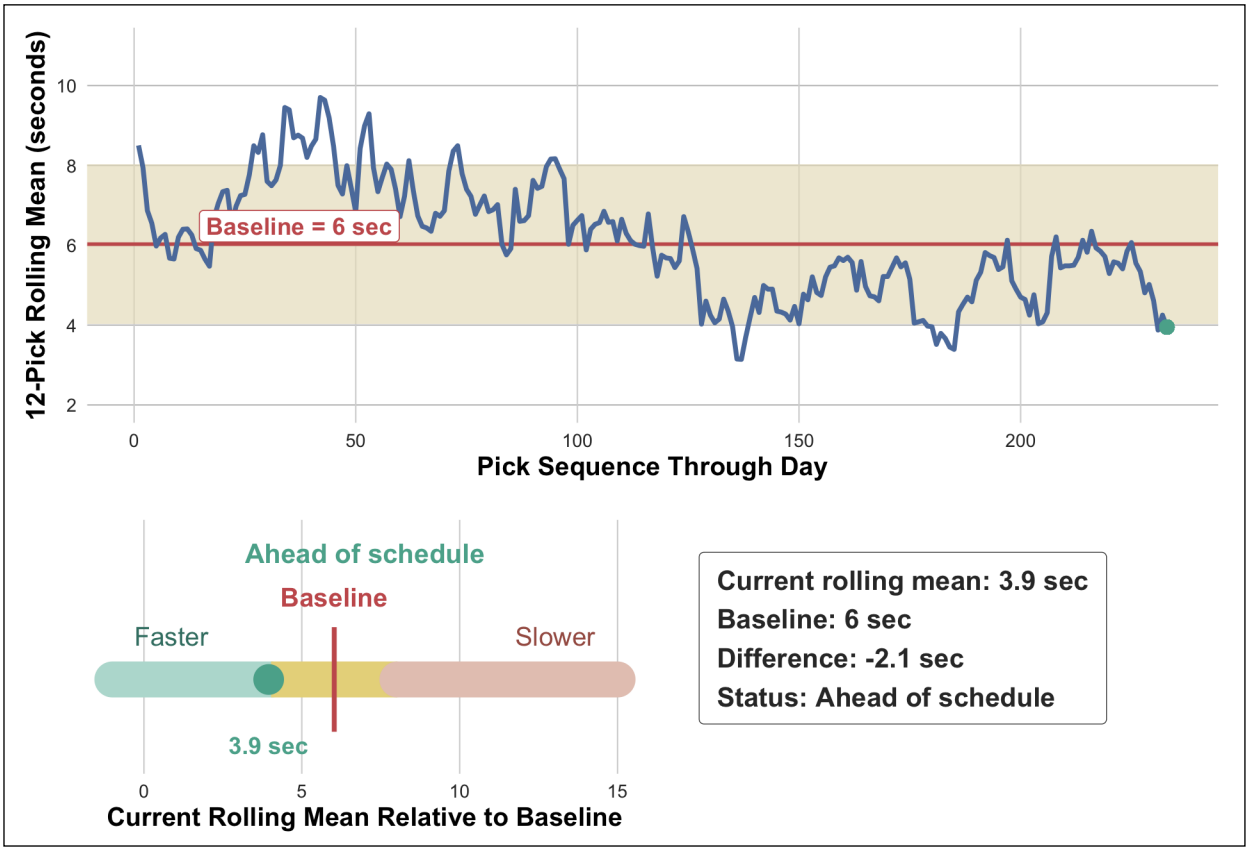


Figure 3. Real-Time Rolling Pick Efficiency Tracker for Day 31

This figure tracks pick efficiency throughout Day 31 using a 12-pick rolling mean of package search times. The top panel shows how average pick time changes across the sequence of deliveries, while the lower panel summarizes current performance relative to the baseline. By the end of the route, the rolling mean reached 3.9 seconds, indicating performance ahead of the baseline target. This same Day 31 route is broken down by package type in **Figure 4** and by bag in **Figure 5**.

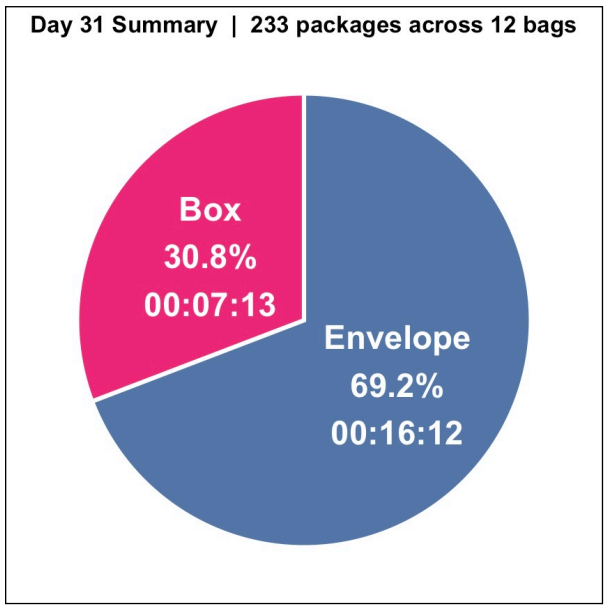


Figure 4. Distribution of Package Pick Time by Package Type on Day 31

This pie chart shows how the total time spent searching for packages on Day 31 was distributed between boxes and envelopes. Envelopes accounted for the majority of search time (69.2%), while boxes accounted for 30.8%. The labels inside each section display the percentage of total pick time and the cumulative time spent locating each package type during the route. This day-level workload split mirrors the broader package-type pattern first introduced in **Figure 1**.

Building Trends, Day by Day

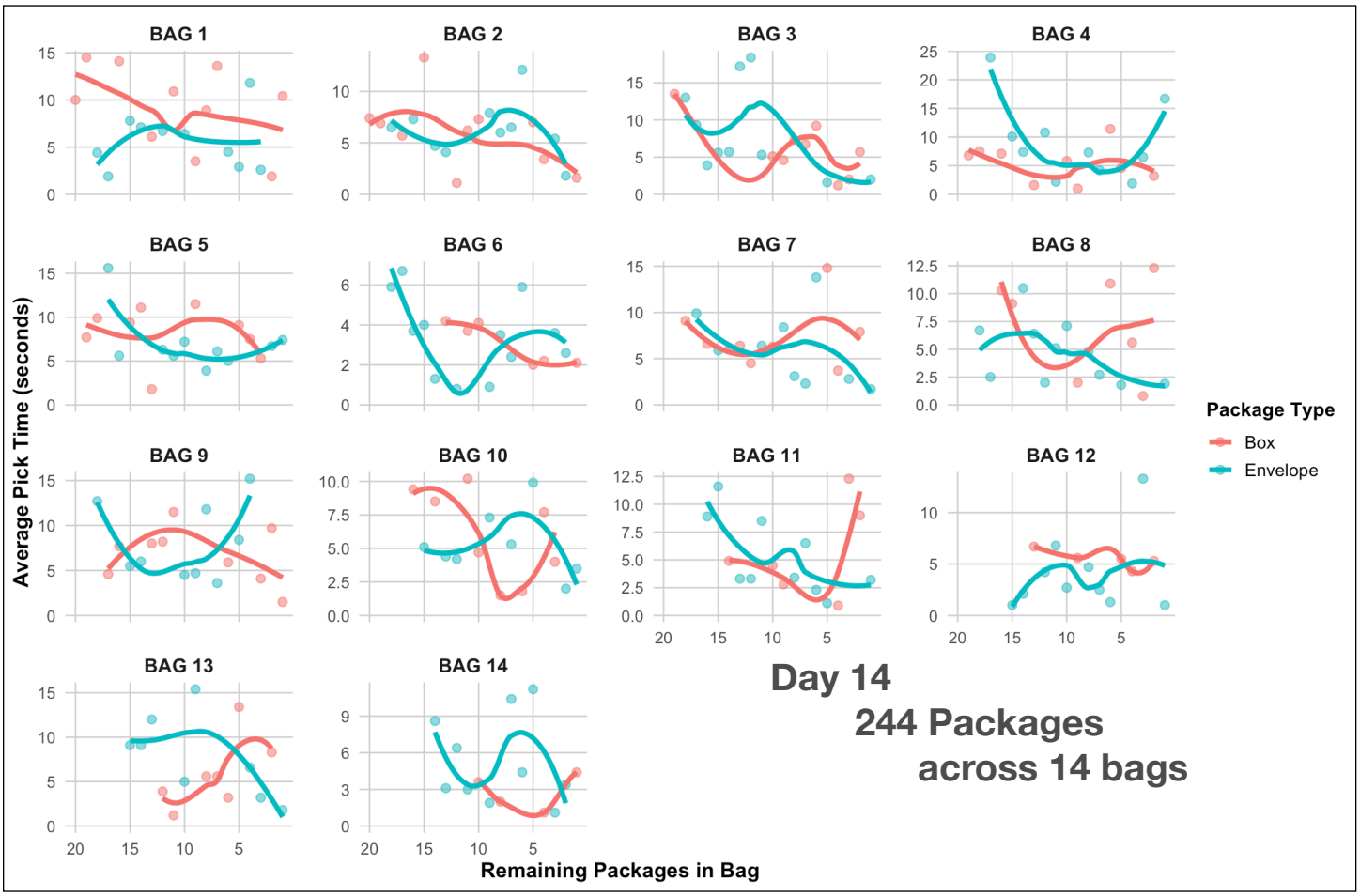


Figure 5. Bag-Level Pick Time Trends on Day 31

This figure shows pick time patterns for each delivery bag on Day 31. Each panel represents one bag and plots the relationship between the number of remaining packages and the time required to locate a package. Individual points show observations, while the smoothed lines summarize the trends for boxes and envelopes. Across the panels, pick times tend to be higher and more variable early in a bag, then faster and more stable as packages are removed. This bag-level pattern is generalized across the full month in **Figure 6**.

Full Month Review

The following visualizations expand the analysis from a single day to the entire month of January. By aggregating results across hundreds of bags and thousands of package picks, broader workflow patterns become clearer and more stable. This larger view allows the daily trends observed earlier to be evaluated at scale, showing how package composition and bag density influence search time across the full dataset.

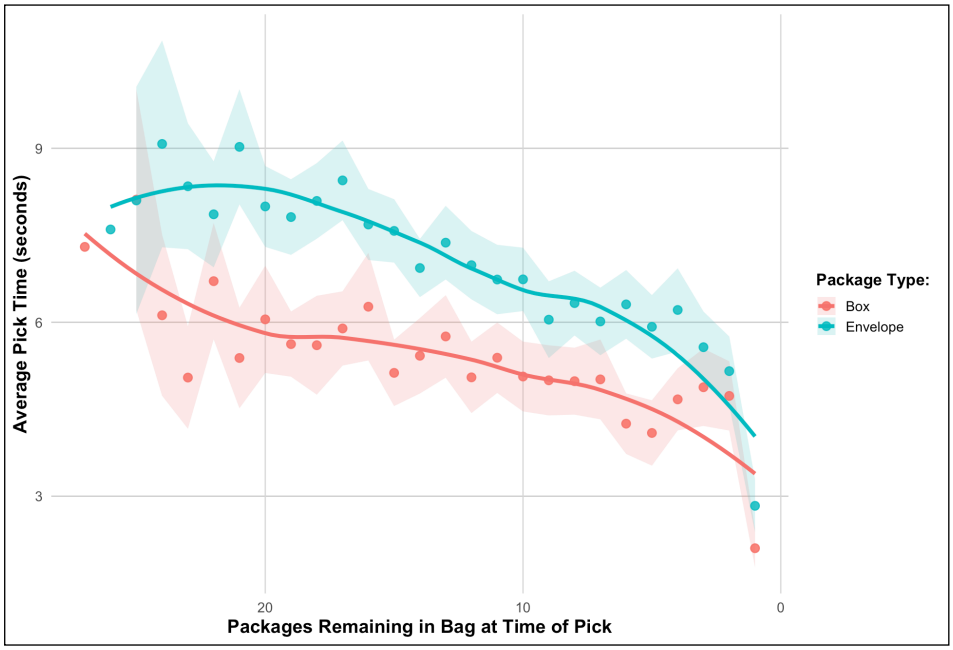


Figure 6. Average Pick Time by Remaining Packages in Bag Across the Full Month

This figure extends the bag-level patterns observed in **Figure 5** by examining the relationship between remaining packages and pick time across the full month of January 2026. Each point represents the mean pick time at a given remaining-package level, separated by package type, while the smoothed curves show the overall efficiency trend as bags empty. Across the dataset, early bags tend to have slightly higher pick times, while later bags tend to be faster. Together, these results reinforce the earlier observation that search efficiency improves as bag density decreases.

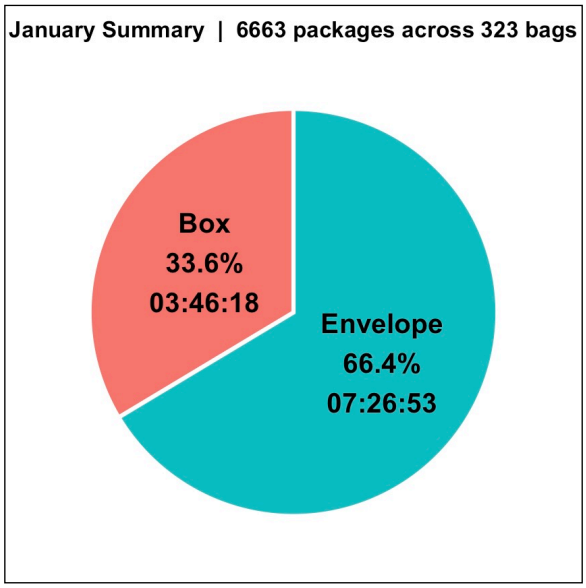


Figure 7. Share of Total Pick Time by Package Type Across January 2026

This pie chart summarizes how total package search time was distributed between boxes and envelopes across the month of January. Rather than focusing on package counts, this figure emphasizes the total time spent locating each package type and shows where the majority of search effort occurred across the workflow. Envelopes account for the larger share of total pick time, reinforcing the package-type differences first introduced in **Figure 1** and echoed again in **Figure 4**.

Report Summary

This project explored how package organization affects search efficiency during an Amazon delivery route. Rather than using a preexisting dataset, I built this project around a workflow I know firsthand as an Amazon delivery driver. One of the most repeated actions in the job is locating the next package, and although that may seem like a small task, it becomes a major part of the day when repeated hundreds of times. I wanted to understand that process statistically and visually, using data to examine what makes picks faster or slower.

A major reason I chose this topic is that it connects directly to my own experience. I know what it feels like to struggle with a crowded bag, to get lucky and find a package instantly, or to lose time searching through envelopes that all look similar. That made the project feel meaningful from the beginning. I also wanted the report itself to feel thoughtful and readable, almost like a short textbook chapter. Every page and figure was arranged intentionally so that the visuals would flow together and gradually build the story.

I began by explaining the two main package types, boxes and envelopes, because that context is essential for understanding the rest of the project. Boxes are more rigid and easier to organize visually, while envelopes are more flexible, more numerous, and harder to search through. That difference became immediately visible in the violin plot, which showed that boxes tend to have shorter and more concentrated pick times, while envelopes have a wider distribution and require more search effort overall.

From there, I moved into weekly comparisons, which helped show how efficiency changed over time. The mirrored histograms revealed that the pick-time distribution gradually shifted downward across the month, meaning there were more fast picks in later weeks than in earlier ones. This suggested that repetition, familiarity, and improved organization all contributed to better performance.

The daily dashboard and pie chart on Day 31 brought the analysis closer to the rhythm of actual work. The rolling mean tracker showed how performance changed throughout a single route, while the pie chart showed how much of that day's total search effort was spent on boxes versus envelopes. Together, those figures captured both the pace of the work and the uneven workload created by different package types.

My favorite visual was the bag-level panel chart, which displayed trends for each bag on Day 31. Even though every bag looked slightly different, the same pattern appeared repeatedly: early in a bag, when inventory was dense and visually crowded, pick times were higher and more variable; later in a bag, as packages were removed, search times became faster and more stable. The full-month trend chart confirmed that this pattern was not limited to one route. Across the month, average pick times dropped from 7.42 seconds in early bags to 5.11 seconds in late bags, showing that search efficiency improved as bag density decreased.

Overall, this project showed that organization, package type, and bag density all shape delivery efficiency. More importantly, it demonstrated that a familiar real-world workflow can be turned into a compelling statistical story through careful design, thoughtful visual sequencing, and transforming my lived work experience into living, breathing visuals.